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A novel method to simultaneously record spinal cord electrophysiology and electroencephalography signals

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ABSTRACT

The brain and the spinal cord together make up the central nervous system (CNS). The functions of the human brain have been the focus of neuroscience research for a long time. However, the spinal cord is largely ignored, and the functional interaction of these two parts of the CNS is only partly understood. This study developed a novel method to simultaneously record spinal cord electrophysiology (SCE) and electroencephalography (EEG) signals and validated its performance using a classical resting-state study design with two experimental conditions: eyes-closed (EC) and eyes-open (EO). We recruited nine postherpetic neuralgia patients implanted with a spinal cord stimulator, which was modified to record SCE signals simultaneously with EEG signals. For both EEG and SCE, similar differences were found in delta- and alpha-band oscillations between the EC and EO conditions, and the spectral power of these frequency bands was able to predict EC/EO behaviors. Moreover, causal connectivity analysis suggested a top-down regulation in delta-band oscillations from the brain to the spinal cord. Altogether, this study demonstrates the validity of simultaneous SCE-EEG recording and shows that the novel method is a valuable tool to investigate the brain-spinal interaction. With this method, we can better unite knowledge about the brain and the spinal cord for a deeper understanding of the functions of the whole CNS.

1. Introduction

Like in all mammals, the human central nervous system (CNS) includes a brain and a spinal cord. To fully understand the functions of the CNS, we need to unite knowledge about the brain and the spinal cord. Yet, the brain is doubtlessly the favorite of cognitive neuroscience research; the spinal cord, on the other hand, is often ignored. To study the functions of the brain, researchers have developed a wealth of innovative techniques (Raichle, 2000; Sakkalis, 2011; Stam, 2005). Some of them — for example, electroencephalography (EEG) and magnetoencephalography (MEG) — record brain activities directly, whereas others, like positron emission tomography (PET) and functional magnetic resonance imaging (fMRI), invoke metabolic or other physiological changes to indirectly record brain signals. These techniques have contributed substantially to our understanding of the brain and are widely used in almost all domains of cognitive neuroscience (Cameron et al., 2017; Collins et al., 2018; Greene et al., 2001; Huang et al., 2013).

In stark contrast, few techniques have been or can be applied to study the functions of the human spinal cord, except, notably, spinal cord fMRI. Yoshizawa and colleagues performed the first fMRI study in the human spinal cord over twenty years ago (Yoshizawa et al., 1996), followed by a handful of task-based spinal cord fMRI studies (Madi et al., 2001; Moffitt et al., 2005; Stroman et al., 1999) and a few resting-state spinal cord fMRI studies in the last two decades (Barry et al., 2014; Eippert et al., 2017; Kinany et al., 2020; Wei et al., 2010). These interesting studies notwithstanding, spinal cord fMRI studies are still facing some technical challenges, for example, cardiac- and respiratory-related motion artifacts and small cross-sectional dimensions (Harita and Stroman, 2017). Also, fMRI measures blood oxygenation level-dependent (BOLD) signals, which only indirectly reflect spinal neural activities and have a relatively poor temporal resolution.

Consequently, it would be of great interest to cognitive neuroscientists, especially for those working on the somatosensory and motor systems, if human spinal cord electrical activities can be measured

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directly. Some researchers have attempted to record spinal cord potentials with electrodes attached to the skin around the spine (Small et al., 1978) or directly inserted into the epidural space (Berić et al., 1986; Dimitrijevic et al., 1980; Jones et al., 1982). However, the former approach, recording signals only from the skin, suffers from poor signal-to-noise ratio (SNR) due to signal loss induced by the vertebra and spinal column. This problem is less concerning for epidural recordings. For example, Dimitrijevic et al. (1980) proved that signal quality was markedly improved when signals were recorded with electrodes from the epidural space. With improved SNR, some researchers have recorded different components (e.g., P10, N13, and P18) of the cervical somatosensory evoked potentials in the epidural space at the dorsal aspect of the cervical spinal cord (Berić et al., 1986). Nevertheless, recording signals only from the epidural space is unable to reveal how the brain and the spinal cord interact since no cortical signals are simultaneously obtained. Simultaneous recording of electrical signals from the brain and the spinal cord would be a powerful tool to understand how the CNS as a whole implements mental processes and encodes pathological states. For example, simultaneous corticospinal recording can be used to reveal how some chronic pain is jointly encoded in the brain and the spinal cord. However, simultaneous recording is rarely adopted. For a few studies which did record both cortical and spinal electrophysiological signals (Shimoji et al., 1971), the functional relationship between the brain and the spinal cord was not fully investigated. As a result, the neural mechanisms of many behaviors and mental processes still remain only partly understood.

To address these issues, we modified a spinal cord stimulator originally implanted to treat pain from postherpetic neuralgia (PHN) to directly record spinal cord electrophysiological (SCE) activities and combined it with simultaneous EEG recording to reveal how the brain and the spinal cord interact functionally. With high temporal resolution, the simultaneous SCE-EEG recording can track the temporal interaction between the brain and the spinal cord and help understand the functions of the CNS more thoroughly. To validate this novel method and showcase its potential applications, we recruited nine PHN patients implanted with the stimulator and obtained SCE and EEG signals using a classical resting-state study design, i.e., two conditions: eyes-closed (EC) and eyes-open (EO). According to previous resting-state studies (Chen et al., 2008; Kan et al., 2017), we hypothesized that the EEG spectral power of the delta frequency band would be lower in the EC condition than in the EO condition, while the EEG spectral power of the alpha frequency band would be higher in the EC condition than in the EO condition. To explore the relationship between EEG and SCE signals and showcase the extra information that simultaneous SCE-EEG recording could provide, we also performed functional connectivity and causal connectivity analyses between EEG and SCE signals. Note that these implanted stimulators, while modified to record spinal neural activities, will not cause any extra trauma to patients and can still work properly after the completion of the experiment.

2. Materials and methods

2.1. Subjects

Nine patients with PHN were recruited from September 2019 to December 2019 at the Department of Pain Management, the Second Affiliated Hospital of Guangzhou Medical University. The diagnosis of PHN was based on clinical symptoms (including medical history, typical scars, pain severity, and pain types) and was confirmed by pain specialists in the hospital. Exclusion criteria included a history of neurological diseases or dementia, psychiatric disorders, or inability to complete the testing procedure. One patient withdrew from the study due to personal reasons. Therefore, data from 8 PHN patients (5 females) aged 68.1 ± 8.7 years (mean ± SD, range = 53–84 years) were collected in this study (see Table 1 for detailed information). All patients gave their written informed consent and were paid for their participation. The clinical research and application ethics committee of the Second Affiliated Hospital of Guangzhou Medical University approved the procedures.

2.2. Equipment installation

All patients were implanted with a spinal cord stimulator (the implanted electrode locations, confirmed using X-Rays, were shown in Supplementary Fig. 1), a device that delivers weak electrical pulses via a set of electrodes placed on the spinal cord epidural. The weak electrical pulses, stimulating certain segments of the dorsal part of the spinal cord, could block the transmission of nociceptive signals traveling from peripheral sensory nerves to the brain, thus causing pain relief (Jensen and Brownstone, 2019; Jeon, 2012). After implantation, the electrical stimulator generally remains in operation until the end of the treatment (Jeon, 2012). In this study, the spinal cord stimulator, a product of Medtronic (Medtronic, Inc.), consists of two main parts: the *in vivo* implant (Model: 3777/3778) and the *in vitro* control set (Model: 355,531) (Supplementary Fig. 2). The *in vivo* implant mainly includes electrodes and a lead, and the *in vitro* control set includes a lead housing, a cable, a connector plug, and an external neurostimulator. There are eight electrodes at the end of the lead (E1 – E8, *in vivo*), and the other end of the lead (*in vitro*) is placed in the extracorporeal lead housing to be connected to the external neurostimulator by the cable and connector plug.

To ensure that the *in vivo* implant could be used for SCE data collection, we removed the *in vitro* control set and connected the *in vitro* end of the lead with a new lead housing for SCE data collection (the details about how to make a lead housing for SCE data collection were shown in Supplementary Fig. 2). The lead housing for SCE data collection was then connected with a Brain Products system (Brain Amp DC 64-channels), which was also used for EEG data collection. In other words, the eight-electrode SCE data were simultaneously collected with EEG data using the same recording equipment. Please note that we prepared a new *in vitro* lead housing, which was originally used to connect the *in vivo* implant with the external neurostimulator. In this study, the new lead housing was adopted to connect the implant with the Brain Products recording system when used to collect electrophysiological data during our experiment. The modification only involved this new *in vitro* lead housing and its connections and did not involve the *in vivo* implant. As a result, our modification would neither cause any harm to the patients nor destroy the patients' treatment device (as emphasized in Supplementary Fig. 2).

2.3. Experimental procedures and data collection

EEG and SCE signals were recorded simultaneously from all patients. To make sure that patients did not experience severe and prolonged pain during data collection, we scheduled the experiment for 7–10 days after the spinal cord stimulation (SCS) implantation, at which time patients would not feel pain even if the stimulator is turned off. Before the experiment, we replaced the lead housing for SCS treatment with the lead housing for SCE data collection. After the experiment, we changed the setting back and resumed the electrical stimulator for all patients. Throughout the entire experiment, the experimenters were disinfected and wore disposable medical gloves.

During the experiment, patients were seated in a comfortable chair in a silent, temperature-controlled room. The experiment consisted of two sessions: an EC session and an EO session. In the EC session, patients were asked to close their eyes, keep relaxed but not fall asleep for 3 min. In the EO session, patients were asked to look at a fixation point on a computer screen and keep relaxed for 3 min. The order of the two sessions was counterbalanced across subjects. In both sessions, EEG data were collected from 21 AgCl electrodes (FPz, FP1, FP2, Fz, F3, F4, Cz, C3, C4, Pz, P3, P4, Oz, O1, O2, P7, P8, T7, T8, F7, and F8) placed on the scalp according to the International 10–20 system (pass band: 1–1000 Hz; sampling rate: 5000 Hz). FCz was used as the recording

Table 1
Demographic and clinical information of PHN patients.

Patients	Gender	Age (years)	Pain duration	Shingles distribution	SCS electrode position
#1	female	65	> 1 month	Left shoulder and neck	C2, C3, C4, C5
#2	female	61	> 2 months	Right abdomen	T4, T5, T6, T7
#3	male	72	> 2 months	Left abdomen and back	T4, T5, T6, T7
#4	female	84	1 month	Left chest and back	T2, T3, T4, T5
#5	male	70	3 weeks	Left waist abdomen	T9, T10, T11, T12
#6	female	65	> 1 month	Left armpit and inner arm	C7, T1, T2, T3
#7	female	75	1 month	Left chest and back	T2, T3, T4, T5
#8	male	53	> 1 year	Right chest and back	T5, T6, T7, T8

reference, and the impedances of all electrodes were kept lower than 10 kΩ. SCE data were collected from eight electrodes (E1–E8) placed on the surface of the spinal cord epidural (Supplementary Fig. 1). The parameters for SCE recording were identical to those for EEG data collection.

2.4. Data preprocessing

EEG and SCE data were preprocessed using EEGLAB (Delorme and Makeig, 2004), an open-source toolbox running in the MATLAB (Mathworks, USA) environment. The data were first bandpass filtered from 1 to 100 Hz, and then notch filtered from 48 to 52 Hz. Artifacts of eyeblinks, movements, and heartbeats were removed using an independent component analysis algorithm (Delorme and Makeig, 2004). For each subject and each condition, continuous EEG/SCE data (3 min) were segmented into 90 two-second epochs without overlap between consecutive epochs, and epochs contaminated with gross artifacts were manually excluded for further analysis. EEG signals were not re-referenced, that is, the reference was still FCz, whereas SCE signals were re-referenced to the common average reference of all eight SCE electrodes. This operation would ensure that EEG and SCE signals were not influenced by each other due to the choice of a common physical reference. Raw SCE traces and their artifact-corrected versions from two representative patients were shown in Supplementary Fig. 3.

2.5. Spectral analysis

To extract the spectral information of EEG and SCE signals, we fast Fourier transformed both signals to the frequency domain for each subject and each condition, yielding the spectra ranging from 1 to 100 Hz (Li et al., 2020; Zhang et al., 2019). The spectra of each frequency were normalized by dividing them by the summated spectral power from 1 to 100 Hz. For all frequency bands (delta: 1–4 Hz, theta: 4–8 Hz, alpha: 8–13 Hz, beta: 13–30 Hz, low gamma: 30–50 Hz, and high gamma: 50–100 Hz), the summated power within their respective spectral ranges was calculated for both EC and EO conditions. For EEG signals, scalp topographies of spectral power at different frequency bands were computed by spline interpolation (Fig. 1A). For SCE signals, the spectral power of all electrodes was vertically displayed to show their spatial distributions for different frequency bands (Fig. 1B).

2.6. Machine learning-based classification

To assess whether EEG and SCE spectral power in delta- and alpha-band oscillations was able to predict the behaviors associated with EC and EO, we built nine support vector machine (SVM) classifiers with alpha-power, delta-power, and the combination of alpha- and delta-power of EEG, SCE, and the combination of EEG and SCE signals as predictors of the EC and EO conditions (Barry and De Blasio, 2017). We adopted the leave-one-out cross-validation method (LOOCV) to assess the performance of SVM classifiers. Specifically, the spectral power of delta and alpha-band oscillations was extracted for all single epochs, the extracted spectral power of all but one epoch was used to train SVM classifiers using the linear kernel function based on the libSVM toolbox (Chang and Lin, 2011), and the remaining epoch was used to test the classification performance. The whole process was repeated for as many times as the number of epochs, each time with a different epoch as the test set. The classification performance was quantified as the prediction accuracy, namely, the proportion of epochs in which the behaviors (EC and EO) were correctly predicted.

2.7. Corticospinal connectivity analysis

To determine whether EEG and SCE signals were functionally connected, we computed phase- and power-based connectivity between these two types of signals. For the phase-based connectivity, we calculated the weighted phase lagged index (wPLI) (Vinck et al., 2011) between EEG and SCE signals in the EO and EC conditions. WPLI is a phase synchronization index less sensitive to noise and the volume conduction problem (Li et al., 2020; Vinck et al., 2011). It is defined as:

$$wPLI(f) = \frac{M^{-1}\sum_{m=1}^{M}\left|imag\left(X_a(f)\cdot X_b^*(f)\right)\right|\cdot sign\left(\varphi_{f,a,m}-\varphi_{f,b,m}\right)}{M^{-1}\sum_{m=1}^{M}\left|imag\left(X_a(f)\cdot X_b^*(f)\right)\right|} \quad (1)$$

where f is the frequency of interest, M is the number of trials, $X_a(f)$ is the Fourier coefficients for signals at sensor a, $X_b^*(f)$ is the complex conjugate of $X_b(f)$, $imag()$ takes the imaginary part of a complex number, $sign()$ is the sign function, and $\varphi_{f,a,m}$ is the phase angle of frequency f from sensor a on trial m. In this study, a and b stand for EEG and SCE signals. The value of wPLI ranges from 0 to 1, with higher values indicating higher connectivity.

For the power-based connectivity, we calculated Spearman's rank-order correlation between EEG power changes (EC-EO) and SCE power changes. Specifically, for all six frequency bands (defined in ***2.5 Spectral analysis***), the power in the EO condition was first subtracted from that in the EC condition, and then the power differences were rank-ordered and used to compute the Spearman's rank-order correlation coefficient between EEG and SCE.

2.8. Causal connectivity analysis

To explore the causal relationship between EEG and SCE signals, lagged partial directed coherence (lPDC), which was estimated based on an extended multivariate autoregressive (eMVAR) model, was calculated using the eMVAR toolbox (Baccalá and Sameshima, 2001; Faes et al., 2013). Specifically, EEG and SCE epochs were first down-sampled to 200 Hz and then modeled using the following Equation:

$$\begin{bmatrix} X_1(t) \\ X_2(t) \end{bmatrix} = \sum_{r=1}^{p} \boldsymbol{B}(r)\left(\begin{bmatrix} X_1(t-r) \\ X_2(t-r) \end{bmatrix}\right) + \begin{bmatrix} u_1(t) \\ u_2(t) \end{bmatrix} \quad (2)$$

where $[X_1(t), X_2(t)]$ represents the EEG and SCE signals; $[u_1(t),\ u_2(t)]$ is uncorrelated Gaussian noise representing the model residuals; $\boldsymbol{B}(r)$ is the MVAR coefficients, which describe the dependence of $X(t)$ on $X(t-r)$ ($r = 1, 2, \ldots, p$) while excluding the coefficients related to the instantaneous effects from the desired spectral causality measure (i.e., $r \neq 0$); and p, the largest number of time points prior to t used to

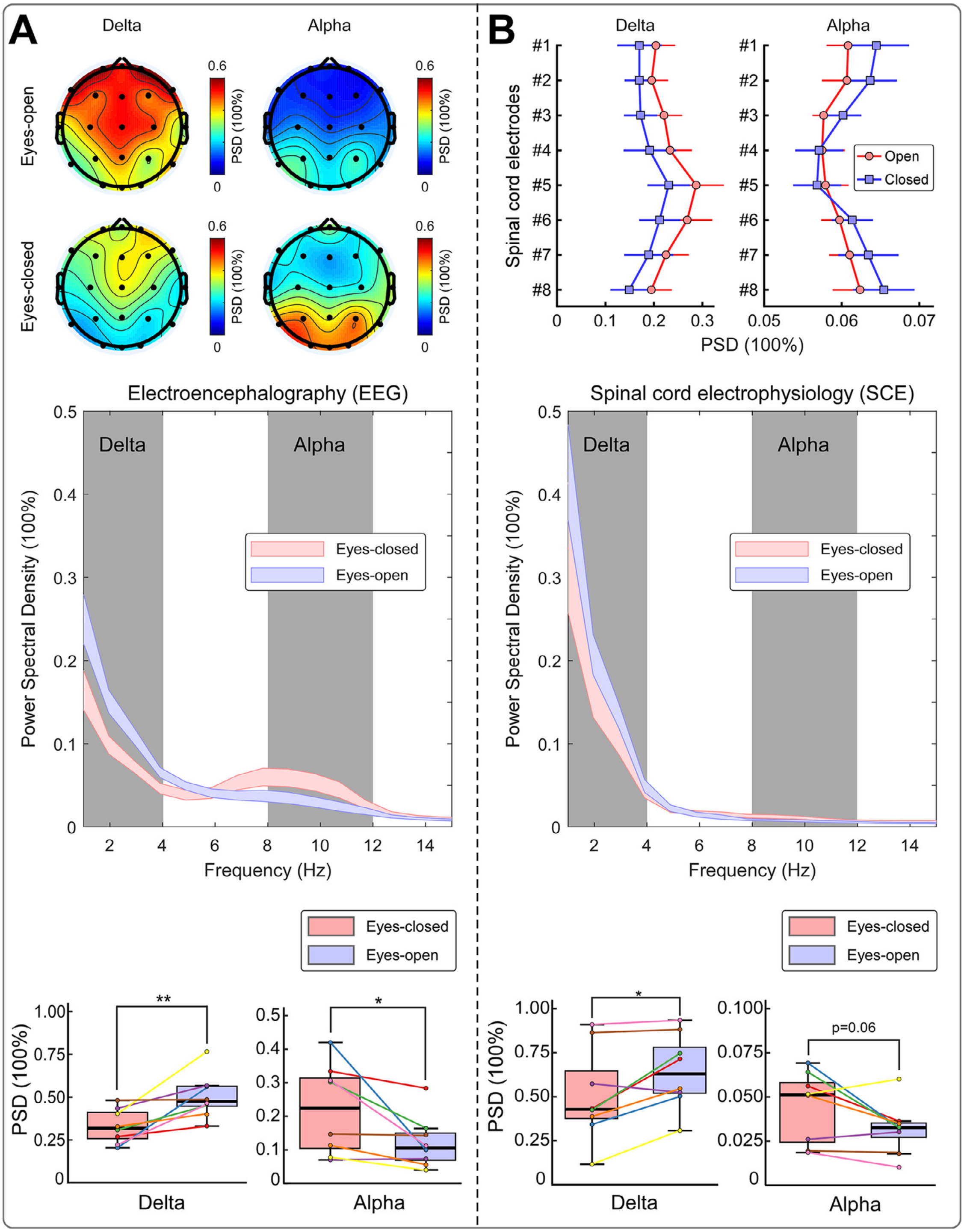

Fig. 1. The comparison of EEG (A) and SCE (B) spectral power between eyes-closed (EC) and eyes-open (EO) conditions. (A) The EEG spectral power in the delta frequency band, which showed a topographical distribution maximal in the frontal-central region of the brain, was significantly lower in the EC condition than in the EO condition. In contrast, the EEG spectral power in the alpha frequency band, which was maximally distributed at the occipital region of the brain, was significantly higher in the EC condition than in the EO condition. (B) The SCE spectral power in the delta frequency band was also significantly lower in the EC condition than in the EO condition. The SCE spectral power in the alpha frequency band was marginally higher in the EC condition than in the EO condition. In EEG and SCE boxplots, the PSD values of each subject in EC and EO conditions are marked with colored dots, and results from the same subject are connected with lines. Different colors represent different subjects, and the same color represents the same subject. Please note that the displayed EEG and SCE spectra are the summated values from all EEG and SCE electrodes, respectively. *: $p<0.05$, **: $p<0.01$.

model $X(t)$, is determined by the performance of the model assessed by Akaike Information Criterion. The lPDC value, i.e., $|C_{i\leftarrow j}(f)|$, is defined as follows:

$$\left|C_{i\leftarrow j}(f)\right| = \frac{B_{ij}(f)}{\sqrt{\sum^{k} \frac{1}{\sigma_{k^2}}\left|B_{kj}(f)\right|^2}} \tag{3}$$

where $B_{ij}(f)$ is the Fourier transformation of the MVAR coefficient, i.e., $\boldsymbol{B}(r)$ in Eq. (2); and σ_k represents the standard deviation of the model residuals. The lPDC value is in the interval of [0 1], where 0 stands for the absence of an influence of the source j on the target i, and 1 stands for a linearly predictable target i from the source j.

2.9. Statistical analysis

For spectral analysis, we used the paired-samples t-tests to assess the difference of spectral power between EC and EO conditions in all frequency bands (delta, theta, alpha, beta, low gamma, and high gamma) for EEG and SCE signals, respectively.

For machine learning-based classification, the prediction accuracy of the nine classifiers was first tested against the chance level (50%) using the one-sample t-test and then evaluated by the two-way repeated-measures analysis of variance (ANOVA) to assess possible effects of "signal type" (three levels: EEG, SCE, and EEG & SCE), "feature type" (three levels: delta-power, alpha-power, delta- & alpha-power), and their interaction. Planned contrasts using the paired-samples t-test were also conducted to test whether prediction accuracy could be enhanced by including delta-power and alpha-power in the classifiers or including EEG and SCE signals in the classifiers. Specifically, first we tested whether "delta- & alpha-power" classifiers had more accurate predictions than "delta-power" or "alpha-power" classifiers in EEG, SCE, and EEG & SCE signals, respectively; then we tested whether "EEG & SCE" classifiers achieved better prediction accuracy than "EEG" and "SCE" classifiers in delta, alpha, and delta & alpha frequency bands, respectively.

For corticospinal connectivity analysis, we used the paired-samples t-test to assess the difference of wPLI between the EC and EO conditions and test whether the Spearman's rank-order correlation between EEG and SCE was significant or not.

For causal connectivity analysis, we used the paired-samples t-test to assess the difference of lPDC values (from EEG to SCE and from SCE to EEG respectively) between EC and EO conditions to determine whether electrophysiological signals in the spinal cord were regulated by brain activities or vice versa.

3. Results

3.1. Spectral power of EEG and SCE

The spectrogram suggested that EEG spectral power differences between the EC and EO conditions were mainly in the delta and alpha frequency bands (Fig. 1A and Supplementary Fig. 4). The scalp topographies showed that delta-power had a spatial distribution maximal in the frontal-central region of the brain and that alpha-power was maximally distributed in the occipital area of the brain (Fig. 1A). Delta-power in EEG was significantly lower in the EC condition than in the EO condition (EC: 0.33±0.04, mean ± standard error, the same hereinafter; EO: 0.50±0.05; $t_{(7)} = -3.621$, $p = 0.008$, Cohen's $d = -1.496$), while alpha-power in EEG was significantly higher in the EC condition than in the EO condition (EC: 0.22±0.05; EO: 0.12±0.03; $t_{(7)} = 2.525$, $p = 0.040$, Cohen's $d = 0.938$). No significant power differences were observed in other frequency bands (see Supplementary Fig. 5 and Supplementary Table 1 for details).

The spectrogram of SCE showed a dominant spectral power at low frequencies, e.g., at the delta frequency band, which also displayed remarkable differences between the EC and EO conditions (Fig. 1B and Supplementary Fig. 4). Statistical comparisons revealed that the delta-power in SCE was significantly lower in the EC condition than in the EO condition (EC: 0.51±0.09; EO: 0.64±0.07; $t_{(7)} = -2.991$, $p = 0.020$, Cohen's $d = -0.579$). Compared to the EO condition, the alpha-power in SCE was higher in the EC condition, but slightly missed the conventional significance level even though the effect was of medium size (EC: 0.04±0.01; EO: 0.03±0.01; $t_{(7)} = 2.193$, $p = 0.063$, Cohen's $d = 0.725$). Moreover, significant EC vs. EO power differences were found in the beta (EC: 0.11±0.02; EO: 0.07±0.02; $t_{(7)} = 2.708$, $p = 0.030$, Cohen's $d = 0.592$), low gamma (EC: 0.10±0.02; EO: 0.07±0.02; $t_{(7)} = 3.188$, $p = 0.015$, Cohen's $d = 0.439$), and high gamma frequency bands (EC: 0.18±0.04; EO: 0.13±0.03; $t_{(7)} = 2.745$, $p = 0.029$, Cohen's $d = 0.485$). No significant difference between the EC and EO conditions was observed in the theta frequency band (see Supplementary Fig. 6 and Supplementary Table 2 for details).

3.2. Machine learning-based classification

As showed in Fig. 2, the "delta-power" classifier (0.66±0.05, $t_{(7)} = 3.477$, $p = 0.010$, Cohen's $d = 1.229$), "alpha-power" classifier (0.69±0.06, $t_{(7)} = 3.064$, $p = 0.018$, Cohen's $d = 1.083$), and "delta- and alpha-power" classifier (0.74±0.05, $t_{(7)} = 4.558$, $p = 0.003$, Cohen's $d = 1.611$) in EEG signals were able to predict EC/EO conditions above the chance level (i.e., 50% prediction accuracy). In contrast, in SCE signals only the "delta- and alpha-power" classifier could predict EC/EO conditions more accurately than the chance level (0.68±0.06, $t_{(7)} = 3.067$, $p = 0.018$, Cohen's $d = 1.084$). If both EEG and SCE signals were fed into the classifiers, the "delta-power" classifier (0.73±0.06, $t_{(7)} = 4.211$, $p = 0.004$, Cohen's $d = 1.489$), "alpha-power" classifier (0.73±0.06, $t_{(7)} = 3.934$, $p = 0.006$, Cohen's $d = 1.391$), and "delta- and alpha-power" classifier (0.81±0.05, $t_{(7)} = 5.736$, $p = 0.0007$, Cohen's $d = 2.208$) all predicted the EO/EC conditions above the chance level.

Two-way repeated-measures ANOVA showed that the prediction accuracy was modulated by the factor "signal type" ($F_{(2,14)} = 4.467$, $p = 0.028$, partial $\eta^2 = 0.118$), but not by the factor "feature type" ($F_{(2,7)} = 3.250$, $p = 0.069$, partial $\eta^2 = 0.194$) and their interaction ($F_{(4,14)} = 1.614$, $p = 0.198$, partial $\eta^2 = 0.011$). Post hoc multiple comparisons showed that the prediction accuracy of classifiers with EEG and SCE signals was significantly higher than that with only SCE signals ($p = 0.009$).

Descriptively, a better prediction accuracy seemed to be achieved by including EEG and SCE signals or delta- and alpha-powers in the classifier. Indeed, among the nine classifiers we considered, the highest accuracy (0.81±0.05) occurred when EEG and SCE signals, and delta- and alpha-powers were used as predictors. This observation was largely confirmed by planned contrasts. For classifiers with EEG and SCE signals, the prediction accuracy of the "delta- & alpha-power" classifier was significantly higher than that of the "delta-power" classifier ($t_{(7)} = 2.597$, $p = 0.035$, Cohen's $d = 0.918$). Similarly, for EEG classifiers, the "delta- & alpha-power" classifier had higher accuracy than the "delta-power" classifier ($t_{(7)} = -2.409$, $p = 0.047$, Cohen's $d = 0.852$). In addition, for SCE classifiers, the prediction accuracy of the "delta- & alpha-power" classifier was significantly higher than that of the "alpha-power" classifier ($t_{(7)} = 2.490$, $p = 0.042$, Cohen's $d = 0.881$), and marginally significantly higher than that of the "delta-power" classifier ($t_{(7)} = 2.261$, $p = 0.058$, Cohen's $d = 0.799$) (Fig. 2).

For classifiers with delta- and alpha-power, the "EEG & SCE" classifier achieved higher accuracy than the "SCE" classifier ($t_{(7)} = 3.168$, $p = 0.016$, Cohen's $d = 1.120$). This pattern also occurred in classifiers with other signal types. For classifiers with delta-power, the "EEG & SCE" classifier had higher accuracy than the "SCE" classifier ($t_{(7)} = 3.469$, $p = 0.010$, Cohen's $d = 1.226$). For classifiers with alpha-power, the "EEG & SCE" classifier had higher accuracy than the "SCE" classifier ($t_{(7)} = 3.594$, $p = 0.009$, Cohen's $d = 1.271$).

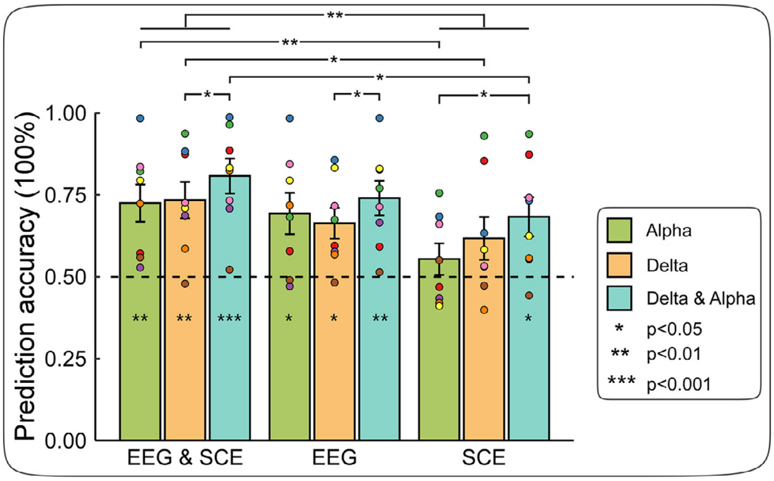

Fig. 2. Prediction accuracy of machine learning-based classifiers in predicting eyes-closed/eyes-open conditions. The prediction accuracy of classifiers with EEG and SCE signals was significantly higher than that with SCE signals. Specifically, the prediction accuracy of "delta-power", "alpha-power", and "delta- & alpha-power" classifiers with EEG & SCE signals was all significantly higher than those with SCE signals. For EEG & SCE signals, the prediction accuracy of the "delta-power", "alpha-power", and "delta- & alpha-power" classifiers was all significantly higher than the chance level (i.e., 50% prediction accuracy). Besides, the prediction accuracy of the "delta- & alpha-power" classifier was significantly higher than that of the "delta-power" classifier. For EEG signals, the prediction accuracy of the "delta-power", "alpha-power", and "delta- & alpha-power" classifiers was all significantly higher than the chance level. Moreover, the prediction accuracy of the "delta- and alpha-power" classifier was significantly higher than that of the "delta-power" classifier. For SCE signals, the prediction accuracy of the "delta- & alpha-power" classifier was significantly higher than the chance level and significantly higher than that of the "alpha-power" classifier. Prediction accuracy of each subject is marked with colored dots. Different colors represent different subjects, and the same color represents the same subject. The dashed line represents 50% prediction accuracy. *: $p<0.05$, **: $p<0.01$, ***: $p<0.001$.

3.3. Corticospinal connectivity

As shown in Fig. 3A, wPLI remained stable across frequency bands for both EC and EO conditions. No statistically significant differences were observed in all frequency bands: (1) delta band: $t_{(6)} = -1.558$, $p = 0.170$, Cohen's $d = -0.589$; (2) theta band: $t_{(6)} = 0.482$, $p = 0.647$, Cohen's $d = 0.182$; (3) alpha band: $t_{(6)} = -0.903$, $p = 0.401$, Cohen's $d = -0.341$; (4) beta band: $t_{(6)} = -2.197$, $p = 0.070$, Cohen's $d = -0.830$; (5) low gamma band: $t_{(6)} = -0.561$, $p = 0.595$, Cohen's $d = -0.212$; (6) high gamma band: $t_{(6)} = -0.185$, $p = 0.859$, Cohen's $d = -0.070$. Taken together, these results suggest that corticospinal phase synchronization may not be modulated by EO/EC behaviors.

Spearman's rank-order correlation revealed a significant positive correlation between EEG and SCE power changes (EC - EO) at the alpha band, Spearman rho = 0.762, $t_{(7)} = 2.881$, $p = 0.028$ (Fig. 3B). In other words, EEG alpha power changes between the EC and EO conditions were monotonically related with SCE alpha power changes. No significant correlations were found in other frequency bands: (1) delta band: Spearman rho = 0.262, $t_{(7)} = 0.665$, $p = 0.531$; (2) theta band: Spearman rho = 0.405, $t_{(7)} = 1.084$, $p = 0.320$; (3) beta band: Spearman rho = −0.143, $t_{(7)} = -0.353$, $p = 0.736$; (4) low gamma band: Spearman rho = −0.643, $t_{(7)} = -2.056$, $p = 0.086$; (5) high gamma band: Spearman rho = 0.048, $t_{(7)} = 0.117$, $p = 0.911$.

3.4. Causal relationship between EEG and SCE

As showed in Fig. 4A, lPDC values from EEG to SCE had two obvious peaks for both EC and EO conditions, one in the delta frequency band and the other in the alpha frequency band. However, lPDC values from SCE to EEG had no marked fluctuation and was almost flat for both EC and EO conditions. Paired-samples *t*-tests indicated that lPDC values from EEG to SCE in the delta frequency band were significantly smaller in the EC condition than in the EO condition (EC: 0.19±0.03; EO: 0.25±0.04; $t_{(7)} = -2.660$, $p = 0.032$, Cohen's $d = -0.642$), while lPDC values from EEG to SCE in the alpha frequency band between the EC and EO conditions were not significantly different (EC: 0.22±0.04; EO: 0.17±0.03; $t_{(7)} = 1.559$, $p = 0.163$, Cohen's $d = 0.466$). For lPDC values from SCE to EEG, no significant differences between EC and EO conditions were observed in both delta and alpha frequency bands (delta: EC: 0.08±0.01; EO: 0.07±0.01; $t_{(7)} = 0.515$, $p = 0.623$, Cohen's $d = 0.314$; alpha: EC: 0.07±0.003; EO: 0.07±0.01; $t_{(7)} = 0.902$, $p = 0.397$, Cohen's $d = 0.454$). These results suggested that electrophysiological signals in the spinal cord probably received top-down regulation from the brain and that this regulation could be reflected in the delta band oscillations.

4. Discussion

In this study, we developed a novel method to simultaneously record SCE and EEG signals and demonstrated its validity using a classical resting-state study design. Specifically, we validated the modified spinal cord stimulator as a recorder of SCE signals, and, with simultaneously recorded SCE and EEG signals, explored the communication between the spinal cord and the brain. For both EEG and SCE, signals in low-frequency bands, including delta (1–4 Hz) and alpha (8–13 Hz), changed between the EC and EO conditions. Combining the delta- and alpha-band power of EEG and SCE predicted the EO/EC conditions more accurately than the delta- or alpha-power alone, suggesting the complementary nature of these two frequency bands. Furthermore, combining EEG and SCE signals achieved better prediction performance than EEG or SCE signals alone, indicating that EC/EO behaviors may be jointly encoded by EEG and SCE signals. Whereas no significant corticospinal phase synchronization differences were observed in any frequency bands, EEG alpha power changes were significantly correlated with SCE alpha power changes. Causal connectivity analysis showed significantly higher lPDC values from the brain to the spinal cord in the delta band in the EC condition compared to the EO condition. Taken together, these results implied that simultaneous SCE-EEG recording is an effective and dependable tool to study spinal cord activity and brain-spinal communication and can help better understand the functions of the CNS.

In EEG, we found that delta-band oscillations in the EO condition increased significantly compared to the EC condition. This finding is in line with many previous studies (Chen et al., 2008; Kan et al., 2017; Zhang et al., 2019). Although functions and mechanisms of sub-band neural oscillations have not been solidly revealed (Buzsáki et al., 2013), it has been a consensus that oscillations have separate and significant functions in the brain instead of just a by-product of the activity of neural networks (Lopes da Silva, 1991). Many researchers believe that delta-band oscillations are related to the activation of cognitive processing (Güntekin and Başar, 2016) and the inhibition of interfering neural activities (Harmony, 2013). In the resting state, EC can be regarded as an "interoceptive" state, characterized by imagination and more sensitive somatosensation (Brodoehl et al., 2015), while EO seems to be an "exteroceptive" state, characterized by attention and ocular motor activity (Marx et al., 2003). In other words, the EO condition requires relatively more cognitive resources than the EC condition. Therefore, delta-power would increase under the relatively high cognitive loading situation where patients were passively exposed to possible visual stimuli in the EO condition.

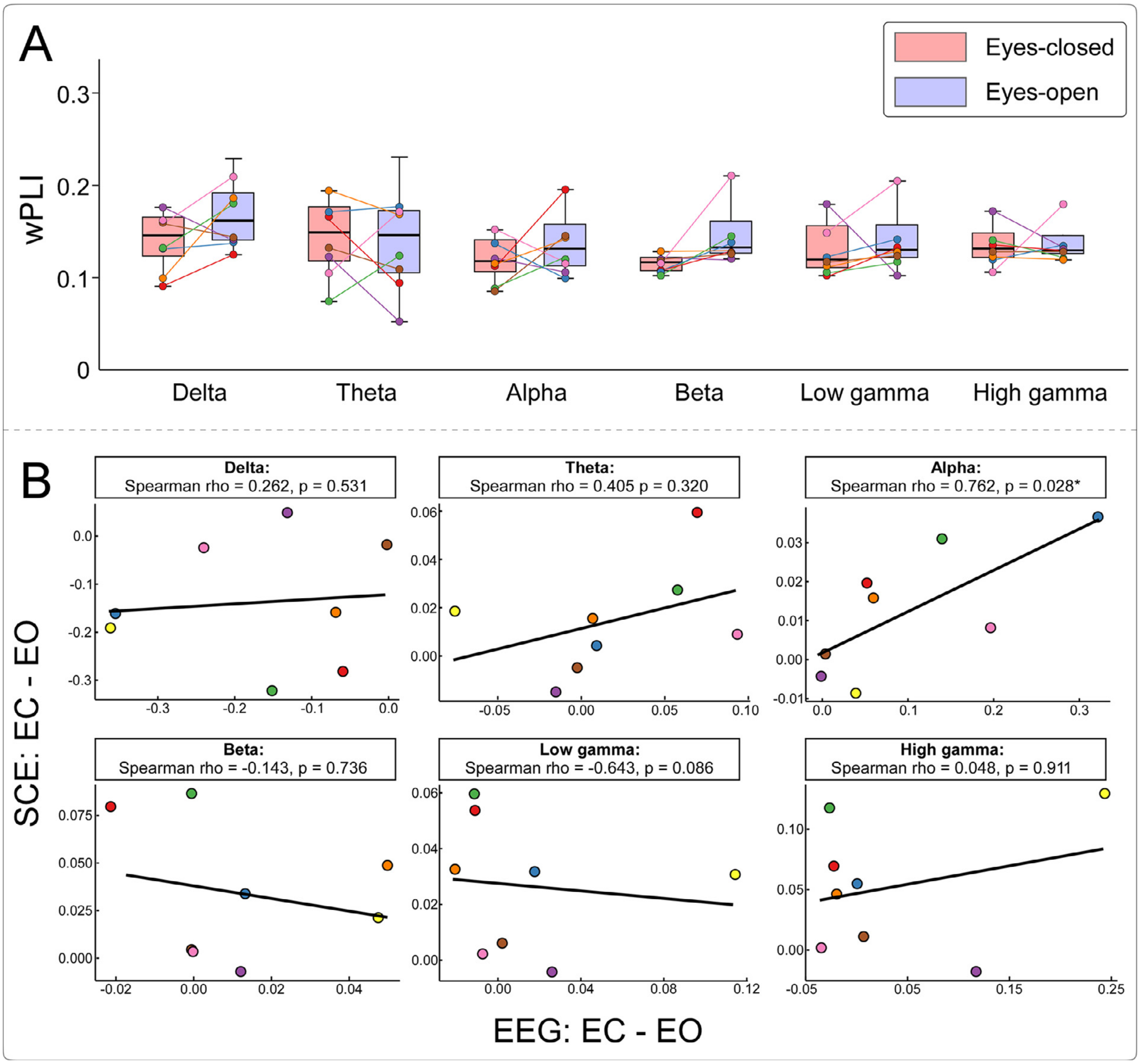

Fig. 3. Corticospinal connectivity between EEG and SCE signals in the EC and EO conditions. (A) Phase-based connectivity between EEG and SCE in six frequency bands. The wPLI values of each subject in EC and EO conditions are shown with colored dots, and results from the same subject are connected with lines. Different colors represent different subjects, and the same color represents the same subject. No significant difference was found between EC and EO conditions in all frequency bands. Please note that data from one subject were removed from the statistical analysis due to their being more than 3 standard deviations away from the mean wPLI across subjects. (B) Power-based connectivity between EEG and SCE in six frequency bands. The x axis is the EEG power difference between the EC and EO conditions (EC - EO); the y axis is the SCE power difference between the EC and EO conditions. In the alpha band, a significant correlation between EEG power differences and SCE power differences was observed. *: $p<0.05$.

Another significant finding in EEG was the larger alpha-power in the EC condition than in the EO condition, which is reminiscent of multitudinous previous studies dating back to almost a century ago (Adrian and Matthews, 1934; Barry et al., 2007; Berger, 1929). Since alpha-band oscillations were first described in Berger's (1929) study, plenty of research has confirmed that simply opening eyes lowers the alpha-power (Barry et al., 2007; Chen et al., 2008; Kan et al., 2017). Recent studies associated the changes in alpha-band oscillations with the inhibition of basic cognitive processes (Jensen and Mazaheri, 2010). For instance, some studies used visual tasks and observed that a power decrease in the alpha-band oscillations was associated with the activation of task-related brain regions and vice versa (Jensen and Mazaheri, 2010; Kelly et al., 2006; Klimesch, 2012). These studies indicated that exposure to visual stimuli during the EO condition could trigger a power decrease in alpha-band oscillations. Our EEG results, highly aligned with these previous studies, therefore testified to the quality of our EEG signals and implied that the recording device itself introduced few, if any, signal distortions. Since EEG and SCE shared the same recording device in the present study, the quality of SCE data would likely be acceptable as well.

In SCE, we found similar patterns of delta- and alpha-band power changes during EC/EO conditions. Delta-power was larger in the EO condition than in the EC condition, while alpha-power was marginally significantly smaller in the EO condition than in the EC condition. These findings indicate that the spinal cord and the brain may encode EC/EO behaviors alike, even though we are not particularly sure whether the functions of alpha and delta frequencies in the spinal cord are similar or identical to those in the brain because of limited existing knowledge. Future studies recording SCE signals may provide a clear answer. Notably, in SCE, we also observed smaller power in beta, low gamma, and high gamma frequency bands in the EO condition than in the EC condition. It should be noted that high-frequency signals are susceptive to muscle

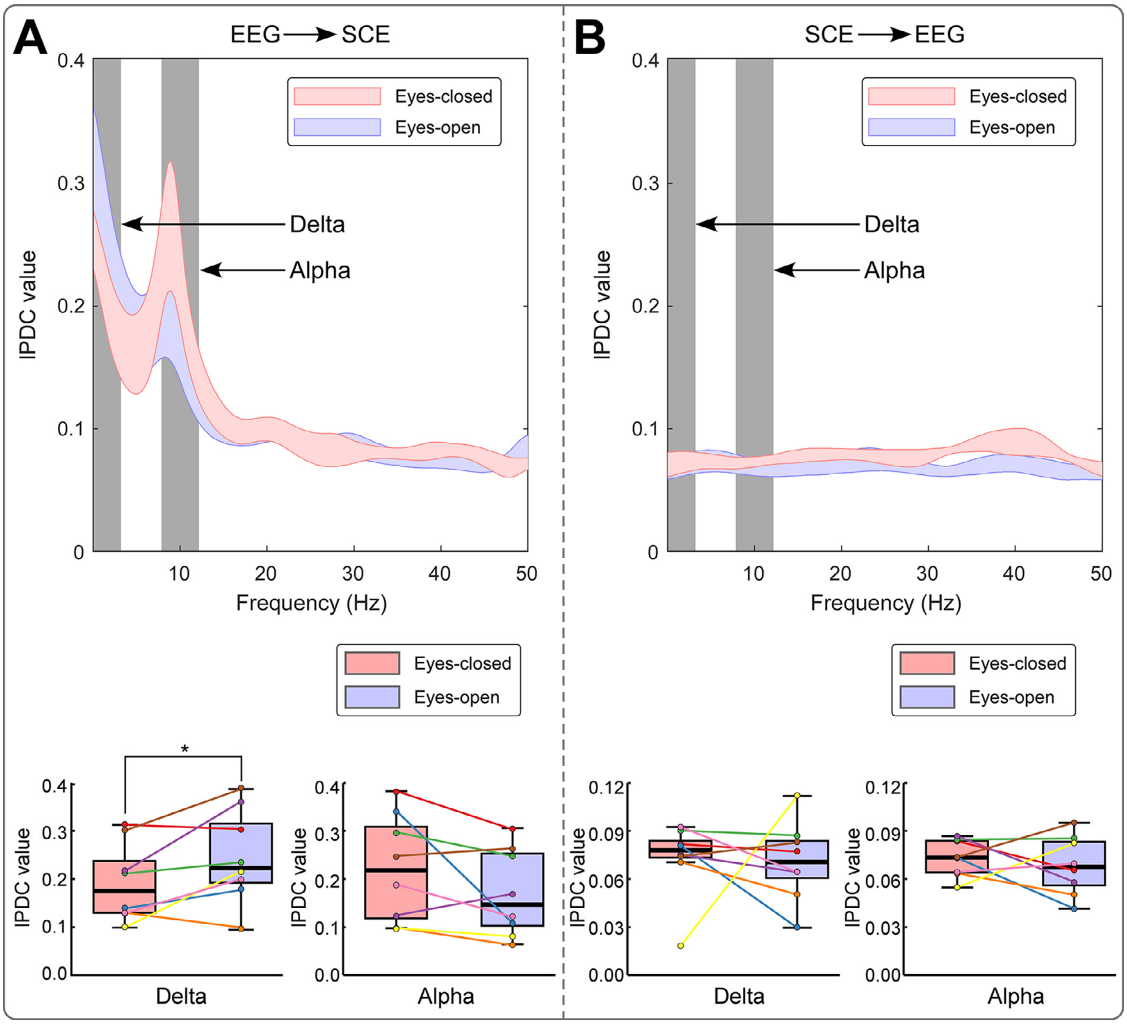

Fig. 4. The causal relationship between EEG and SCE signals. (A) lPDC values from EEG to SCE had two obvious peaks for both EC and EO conditions, one in the delta frequency band and the other in the alpha frequency band. In the boxplots, the lPDC values of each subject in EC and EO conditions are shown with colored dots, and results from the same subject are connected with lines. Different colors represent different subjects, and the same color represents the same subject. In addition, lPDC values in the delta frequency band only were significantly larger in the EO condition than in the EC condition. (B) lPDC values from SCE to EEG had no marked fluctuation and was almost flat for both EC and EO conditions. There were no significant differences in lPDC values between the EC and EO conditions in both delta and alpha frequency bands. *: $p<0.05$.

artifacts (Muthukumaraswamy, 2013) and that the spinal cord is more vulnerable to physiological motion than the brain due to the fact that the spinal cord is covered by many back muscles and moves together with respiration and the flow of cerebrospinal fluid (Stroman et al., 2014). Therefore, we are not able to rule out the possible contribution of muscle artifacts on different patterns of high-frequency power changes in SCE signals, especially when power changes in these frequency bands were not obtained in EEG signals. The detailed mechanisms associated with these high frequencies in SCE signals deserve to be explored in future studies.

To further explore the role of CNS activities in the EC/EO conditions, we fed delta- and alpha-band powers of EEG, SCE, and EEG & SCE signals into the support vector machine to learn and predict EC/EO behaviors. The machine learning-based classification added some important pieces of extra information to the traditional statistical analysis. First, in addition to confirming the results that delta- and alpha-band powers contained information about EC/EO behaviors, the classification analysis indicated that delta- and alpha-band powers may contain complementary information, as improved prediction accuracy could be achieved by combining delta- and alpha-band powers. Moreover, EEG and SCE signals were shown to jointly encode EC/EO behaviors because combining EEG and SCE signals also achieved better prediction performance than only feeding SCE or EEG signals into classifiers. Taken together, these results suggest that EC/EO behaviors are probably encoded in different frequency band powers of EEG and SCE signals.

To determine the functional relationship between the brain and the spinal cord, we also investigated corticospinal connectivity by computing wPLI and Spearman's rank-order correlation coefficients between SCE and EEG signals and the information flow within the CNS using causal connectivity analysis. It turned out that no significant phase synchronization differences between the EO and EC conditions could be observed in any frequency bands. This result implied that corticospinal phase synchronization is probably not modulated by EO/EC behaviors. In contrast, alpha power changes (EC - EO) of EEG signals were significantly correlated with those of SCE signals, suggesting that cortical and spinal alpha power changes tend to remain consistent across individuals. More importantly, there was a significantly higher lPDC from EEG signals to SCE signals in the delta band for the EO condition than the EC condition, but not in other frequency bands. These findings supported the speculation that different frequencies of oscillations favor different types of connection (Harmony, 2013) and indicated a top-down regulation from the brain to the spinal cord during EC/EO conditions. Lower frequencies are believed to facilitate large area information transmission and integration (Buzsaki, 2006). Delta-band oscillations are thus well-suited to encode the information flow from the brain to the spinal cord.

In resting-state studies, subjects are asked to keep relaxed and do practically nothing. Their brains are thus likely to only process visual, auditory, and somatosensory information from the environment and random thoughts which subjects happen to have at the moment. Our findings provided evidence that, in the resting-state, the spinal cord received top-down regulation from the brain. Therefore, the top-down regulation from the brain may thus reflect the brain's attempt to inhibit the somatosensory information, which is probably dominated by visual information when subjects' eyes are open. However, the exact role of the delta band in the brain-spine information flow could not be ascertained in this study alone due to the lack of sufficient existing knowledge. Future studies with experimental tasks (e.g., performing/planning a motor task or receiving tonic/phasic sensory stimulation) are required to improve our understanding of the brain-spine connection. Crucially, these findings also suggested that the simultaneous recording of SCE and EEG may be able to reveal the causal relationship between the brain and the spinal cord.

This study has important implications. First and foremost, apart from recording epidural electrophysiological signals (Berić et al., 1986; Dimitrijevic et al., 1980; Jones et al., 1982), we recorded EEG signals simultaneously, thus providing a valuable tool to investigate the brain-spinal interaction. The spinal cord is in charge of transmitting information from the body to the brain and vice versa. Although only available to a specific population at present, SCE could reveal the spinal mechanism in somatosensory and motor processing, and, combined with simultaneous EEG recording, help better understand the communication between the brain and the spinal cord under various behavioral tasks, at least in certain patients whose SCE signals could be obtained. For example, discovering neuro-markers of pain has been a hot topic recently (Davis et al., 2017; Hu and Iannetti, 2019; Mouraux and Iannetti, 2018; Su et al., 2019), but most of these studies only make use of brain signals. With simultaneous SCE-EEG recording, we can incorporate spinal signals as well and explore the interaction between the spinal cord and the brain in pain conditions. Moreover, much like implanted electrodes in the brain, SCE recording may also help understand more conventional outcomes applicable to healthy subjects, for example, spinal cord fMRI and spinal surface recording (Hubli et al., 2019). In fact, Cioni and Meglio (1986) have checked the consistency of epidural and surface recordings and found that epidural recordings of spinal cord potentials evoked by segmental volleys are similar to those obtained from the skin. Coupled with EEG, simultaneous EEG-SCE recording can further test the reliability and validity of simultaneous EEG and spinal surface recording in future studies.

Albeit its promising theoretical and practical implications, this study has several major limitations. The first is the small number of subjects (i.e., eight). This was due to two main difficulties in recruiting subjects: (1) SCE recording relied on an invasive and implanted stimulator and, as far as this study is concerned, could only be obtained from patients who had received SCS treatment for pain; (2) PHN is often diagnosed in the elderly (Stankus et al., 2000), and very few PHN patients satisfied our subject selection criteria. The problem of the small sample size could lead to insufficient statistical power to detect, for example, the alpha-power difference between the EC and EO conditions in SCE, which slightly missed the conventional significance level in this study ($p = 0.063$). To partly mitigate this insufficient power problem, we did not correct p values reported above. The conclusions of the present study should thus be considered tentative and require further confirmation in future studies. Second, all subjects in this study were PHN patients. Even though we had to recruit patients implanted with the spinal cord stimulator to collect SCE signals, characteristics of EEG and SCE in these patients might have been altered by neuropathic pain. Consequently, some of our findings may not be generalizable to other patients or healthy individuals. Nevertheless, it is important to note that some findings (e.g., alpha power increases in the EC condition as compared with the EO condition) are the same as those from healthy subjects. The third limitation of note is that each subject's spinal cord stimulator was not aligned and only covered limited spinal segments according to each one's individual medical condition. Importantly, aligned recorders and full spinal cord coverage would also require a much larger subject number and long-term data acquisition process. Finally, we only collected resting-state electrical signals. To fully reveal the mechanisms of somatosensory/motor processing in the spinal cord and brain-spinal communication, future studies should include somatosensory/motor tasks while simultaneously collecting SCE and EEG signals.

5. Conclusions

In the present study, we developed a novel method to simultaneously record SCE and EEG signals and validated its performance using a classical resting-state study design. Specifically, we provided a full characterization of the SCE and EEG spectra and demonstrated that the spectral power of SCE and EEG signals was able to predict EC/EO behaviors. Additionally, causal connectivity analysis suggested a top-down regulation from the brain to the spinal cord, and this regulation was reflected in the delta-band oscillations. Altogether, this study demonstrated the validity of simultaneous SCE-EEG recording and showed that the novel method is a valuable tool to investigate the brain-spinal interaction .

Data and code availability

Data are available via reasonable requests to the corresponding author.

Author contributions

Conception and design of the study: FXW, QZ, LW, XLX, and LH. Data acquisition and analysis: FXW, LPY, LBZ, QZ, QJG, JBZ, DYL, CYL, XYL, XLX, LW, XLX, and LH. Drafting the manuscript: FXW, LBZ, LPY, YXZ, LW, and LH. All authors reviewed and accepted the final draft of the manuscript.

Declaration of Competing Interest

All authors declare no competing interests.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.neuroimage.2021.117892.

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